

In Exercises 21 and 22, mark each statement True or False. Justify each answer.

21. a) In some cases, a matrix may be row reduced to more than one matrix in reduced echelon form, using different sequences of row operations.

Ans: By Theorem 1, we know each matrix is row equivalent to one and only one reduced echelon matrix. $\therefore$ The statement is False.

b) The row reduction algorithm applies only to augmented matrices for a linear system.

Ans: On Page 12, Sec 1.2, second paragraph, it is mentioned that the algorithm applies to any matrix, whether or not the matrix is viewed as an augmented matrix for a linear system. $\therefore$ The statement is False.

c) A basic variable in a linear system is a variable that corresponds to a pivot column in the coefficient matrix.

Ans: The statement is True. It is mentioned on Page 18 just below the system of equations (4).

d) Finding a parametric description of the solution set of a linear system is the same as solving the system.

Ans: The statement is True. On Page 19, under the Section Parametric Descriptions of Solution Sets, it is mentioned in the first paragraph.

e) If one row in an echelon form of an augmented matrix is $[0 \ 0 \ 0 \ 5 \ 0]$, then the associated linear system is inconsistent.

Ans: By Theorem 2, a linear system is inconsistent if and only if an echelon form of the augmented matrix has at least one row of the form $[0 \ \dots \ 0 \ b]$ with b non-zero.

$\therefore$ From the given information, we cannot conclude whether the associated linear system is consistent or inconsistent.

22. a) The echelon form of a matrix is unique.